

Contents

Introduction	1
1 Background and Related Work	3
1.1 Single-object tracking	3
1.2 From Siamese trackers to transformer trackers	5
1.3 Template and Search Crop Notation	6
1.4 Evaluation metrics and common benchmarks	7
1.5 Motivation for temporal memory	9
2 OTrack Baseline	11
2.1 OTrack Tracker	11
2.2 ViT Backbone	12
2.3 Candidate elimination	14
3 Proposed Memory-Augmented OTrack	17
3.1 Memory-token design motivation	17
3.2 Direct transformer-based memory update	17
3.3 Limitations of direct memory update	20
3.4 Two-way GRU. GRU-based memory update	21
4 Sequence-Based and Inference-Like Training	24
4.1 Sequence-based training for recurrent memory	24
4.2 Dynamic inference-like search cropping	25
4.3 Truncated Backpropagation Through Time in the training loop	27
5 Auxiliary Memory Supervision	28
5.1 Auxiliary Loss Concept	28

5.2	Sampled Discriminative Filter Loss	28
5.3	DiMP Target-Match Loss	30
5.4	Template Blurring	33
6	Experimental Setup	35
6.1	Dataset and protocol	35
6.2	Model configuration	35
6.3	Evaluation protocol	36
7	Results and Discussion	37
7.1	Quantitative Results	37
7.2	Training Dynamics	38
7.3	Limitations and Future Work	41
	Conclusion	42
	References	44
	References	44
	Summary	47
	Abbreviations	49

Introduction

Modern visual object tracking relies heavily on matching an initial target template to a current search region. While transformer-based architectures have achieved remarkable success in this spatial matching task, they typically process frames independently, lacking an explicit mechanism to accumulate temporal information over the course of a video. Consequently, these models struggle when the target undergoes severe deformation, occlusion, or long-term appearance changes.

To address this limitation, this thesis proposes MemOTrack, a recurrent memory extension of OTrack [13]. OTrack serves as a strong baseline because it jointly processes template and search tokens in a single transformer backbone that can be easily modified. MemOTrack extends this architecture by augmenting the original template-search token sequence with explicit recurrent memory tokens. These memory tokens are continuously refined through self-attention within the Vision Transformer and a Gated Recurrent Unit (GRU) [20] mechanism. This update combines temporal information from previous frames with layer-wise memory refinement, granting the tracker an explicit recurrent state while preserving the core structure of the baseline.

The scope of this work also includes the training changes required for a correct training of a recurrent tracker. As described in the chapter 4 the training pipeline was restructured to use ordered video sequences and inference-like dynamic search cropping. In addition, auxiliary memory losses and template blurring were explored as strategies for encouraging the model to use the newly added memory pathway.

In addition, while Single-Object Tracking (SOT) is utilized as a rigorous test framework for this research, providing well-established datasets, metrics, and baselines, the core contributions of this work are not strictly limited to tracking. The proposed inte-

gration of recurrent memory tokens, layer-wise GRU updates, and dynamic sequence training might also be useful for broader video processing tasks requiring efficient, long-term temporal modeling in Vision Transformers.

The main contributions of this thesis are:

- introduction of recurrent memory tokens into the OTrack ViT backbone;
- implementation of a two-stage GRU memory update;
- rewriting of the training sampler for ordered video sequences;
- implementation of inference-like dynamic search cropping;
- exploration of memory loss heads and template blurring;
- experimental comparison with CE-enabled OTrack baselines at 256 and 384 search resolutions.

1 Background and Related Work

1.1 Single-object tracking

Single-Object Tracking (SOT) is a sequential computer vision problem. In the first frame of a video, a target is identified by a bounding box. In all subsequent frames, the tracker must predict the target’s location without receiving any additional manual annotation. This strict initialization distinguishes tracking from ordinary object detection: identifying the class of the object is insufficient, as the tracker must follow a specific instance. If several visually similar objects are present in the scene, the correct output is the original target, not merely any object of the same category.

The task can be formulated as:

Given frames $I_1, I_2, \dots, I_T$ and the initial bounding box B_1 , a tracker predicts boxes $\hat{B}_t$ for $t > 1$.

Based on how they process this sequence, trackers can be divided into two categories. A stateless **memory-free tracker** can be written purely as a function of the initial template and the current search region, treating each frame independently. In contrast, a **recurrent tracker** additionally utilizes a hidden state carried forward from previous frames to accumulate temporal context. This distinction is central to this thesis, proposed MemOTrack explicitly transitions a stateless baseline (OSTrack) into a recurrent tracker by introducing memory tokens into the backbone.

Early tracking methods were mainly based on hand-crafted appearance and motion models, including mean-shift tracking [1], optical flow-based methods [2, 16], particle filters [3], and template matching approaches [4]. These methods were often efficient and conceptually simple, but they struggled under large appearance changes, occlusion,

background clutter, and scale variation. Around the early 2010s, discriminative correlation filter trackers became one of the dominant families in visual tracking. Methods such as MOSSE [5], CSK [6], KCF [7], and DSST [8] learned online filters that could localize the target efficiently in the frequency domain, while DSST also introduced a separate scale estimation mechanism. Later, more advanced correlation-filter trackers such as ECO [9] improved robustness by using more compact model updates and stronger feature representations. This period is commonly treated as the correlation-filter era of SOT, before deep Siamese and transformer-based trackers became the dominant direction.

Despite these algorithmic advancements, tracking remains an inherently difficult task because the visual appearance of the target is rarely fixed. The target may rotate, deform, change illumination, become blurred by motion, or be partly hidden. Furthermore, the camera itself may move, producing large apparent target displacements between frames. A successful tracking method must therefore balance stability and adaptability: it must not drift to a background distractor, yet it must adapt when the true target changes appearance. Table 1.1 summarizes the most common tracking challenges encountered in SOT benchmarks.

Table 1.1: Common tracking challenges.

Challenge	Example	Why it is difficult
Occlusion	Target is partially hidden	The visible evidence becomes incomplete and the tracker may drift.
Scale change	Target approaches the camera	The size of the target box changes over time.
Deformation	Person, animal, or flexible object changes shape	Appearance no longer matches the initial template exactly.
Background clutter	Similar objects in the scene	The tracker can confuse target and distractor.
Motion blur	Fast motion or camera movement	Fine details disappear from the search crop.
Out-of-view motion	Target leaves and returns	The tracker state may be corrupted during absence.

1.2 From Siamese trackers to transformer trackers

The state of the art in single-object tracking has largely focused on template-to-target matching using neural networks and modern machine learning techniques, where the tracker relies on visual cues from two image regions: a template crop that describes the target appearance and a search crop in which the target has to be localized. A major step in this direction was introduced by SiamFC [10], which formulated tracking as a convolutional Siamese matching problem between an exemplar image and a larger search image. Later, SiamRPN [11] extended this idea by combining Siamese feature extraction with a region proposal subnetwork, allowing the tracker to perform target classification and bounding-box regression. In these early Siamese trackers, the template and search images are processed by two branches with shared convolutional weights, after which the extracted features are compared or fused to estimate the target location.

Later trackers introduced transformer-based components for richer relation modeling between the target and the search region. STARK [12] used an encoder-decoder transformer to model global spatio-temporal dependencies and to directly predict the target bounding box. OSTRack [13] further simplified the tracking pipeline by introducing a one-stream transformer framework. Instead of extracting template and search features independently and then applying a separate relation module, OSTRack jointly processes template and search tokens inside the same backbone, so that feature extraction and relation modeling are unified.

All of these models share a similar high-level tracking structure. They accept a template crop, which represents how the target looks, and a search crop, which contains the region of the current frame where the target is expected to appear. The tracker then extracts visual features from these two inputs and estimates the target position by comparing, fusing, or jointly modeling the template and search representations. The main architectural differences lie in how these features are extracted and how the relation between the template and search region is modeled: earlier Siamese trackers rely mainly on convolution and cross-correlation, while transformer-based trackers use attention mechanisms to allow richer interaction between template and search features.

It is also important to note that feature extraction in modern trackers is often based

on pretrained backbones. These backbones are pretrained on large-scale image or video datasets before being adapted to tracking, because tracking datasets alone are usually not sufficient for learning strong low-level and semantic visual features from scratch. In practice, the pretrained backbone provides general visual representations, while tracker training teaches the model how to use these representations for target localization and target-background discrimination.

Another distinct and highly influential approach worth mentioning was introduced by DiMP [15]. Unlike standard Siamese networks that simply compare the current search region against a fixed initial template, DiMP dynamically optimizes a discriminative target model during inference. By leveraging historical sequence information, it continuously updates a classifier to distinguish the target from background distractors. While DiMP relies on an explicit optimization module rather than a transformer architecture, its success shows that incorporating sequence-level information and online adaptation can significantly improve tracking robustness.

1.3 Template and Search Crop Notation

This thesis follows the standard template-search notation used in many modern single-object trackers. The **template crop** is denoted by T . It is extracted from the first annotated frame and contains the target object we are tracking. It is used to provide the visual description of the object that must be tracked.

The **search crop** at time t is denoted by S_t . It is extracted from the current frame around the expected target location. Usually, a neural-network tracker does not process the full image directly. Instead, it receives a smaller resized crop around the expected target location. This location is usually estimated from the bounding-box prediction in the previous frame.

There are two main reasons why a search crop is used instead of the whole image. First, it reduces computation by allowing the model to process only the region where the target is expected to appear. This is especially important for transformer-based trackers, because standard full self-attention is quadratic in the number of tokens [18]. For example, with 16×16 patches, a 256×256 search crop contains 256 tokens, whereas a full

HD frame would contain more than 8000 tokens. The corresponding attention matrix is therefore much smaller for the search crop than for the full image.

Second, the search crop provides a useful spatial prior. Since the crop is usually centered around the previous target bounding box, it gives the tracker information about where the target was located in the previous frame and what its approximate scale was. In short-term tracking, the target is usually expected to remain near its previous location, so a local search region is often sufficient. The tracker then predicts the new target box inside this resized crop, and the prediction is converted back to the coordinate system of the original image.

In many trackers, the template is cropped only at initialization and remains fixed during the sequence. Other trackers update or maintain an additional target representation during inference. FEAR [14], for example, uses a dual-template representation for object model adaptation, where a dynamic template complements the initial template during tracking.

In contrast, trackers such as OTrack store only the initial template extracted from the first frame and do not maintain an explicit updated visual template during inference. This makes the tracker dependent on the initial target appearance. If the target rotates, deforms, changes illumination, or becomes partially occluded, the initial template may no longer describe the current appearance well. This limitation motivates the temporal memory mechanism developed in this thesis.

1.4 Evaluation metrics and common benchmarks

Because single-object tracking is a widely studied problem, it has established benchmarks and evaluation metrics. These benchmarks make it possible to compare trackers under a common protocol, using the same test sequences and the same localization measures.

The main benchmark used in this thesis is GOT-10k. GOT-10k is a generic object tracking benchmark designed to evaluate how well a tracker generalizes to unseen object classes [17]. It follows a one-shot tracking protocol: the target is specified by a bounding box in the first frame, and the tracker must localize the same object in the remaining

frames. The test annotations are hidden, so final results are obtained through the official evaluation server.

The standard localization measure used in these evaluations is intersection over union (IoU), also called overlap. For a predicted box and a ground-truth box, IoU is the area of their intersection divided by the area of their union. A value of 1 means perfect overlap, while a value of 0 means that the boxes do not overlap.

$$\text{IoU}(B_{\text{pred}}, B_{\text{gt}}) = \frac{|B_{\text{pred}} \cap B_{\text{gt}}|}{|B_{\text{pred}} \cup B_{\text{gt}}|} \quad (1.1)$$

where:

- B_{pred} is the predicted bounding box;
- B_{gt} is the ground-truth bounding box;
- $\cap$ denotes the intersection of the two boxes;
- $\cup$ denotes the union of the two boxes;
- $|\cdot|$ denotes the area of a box region.

GOT-10k reports Average Overlap (AO), SR0.50, and SR0.75 [17]. AO is computed as the average IoU over all evaluated frames:

$$\text{AO} = \frac{1}{N} \sum_{i=1}^N \text{IoU}(B_{\text{pred}}^i, B_{\text{gt}}^i). \quad (1.2)$$

where:

- AO is the Average Overlap score;
- N is the number of evaluated frames;
- i is the frame index;
- B_{pred}^i is the predicted bounding box for frame i ;
- B_{gt}^i is the ground-truth bounding box for frame i ;
- $\text{IoU}(\cdot, \cdot)$ is the intersection-over-union function defined in Equation (1.1).

SR0.50 is the proportion of frames whose overlap is at least 0.50, while SR0.75 is the proportion of frames whose overlap is at least 0.75. SR0.75 is stricter and is more sensitive to precise localization errors. These metrics are useful together because a tracker may localize the target approximately while still producing less accurate bounding boxes.

1.5 Motivation for temporal memory

The motivation for adding temporal memory to a tracker comes from several limitations of using only the initial template and the current search crop. First, the initial template becomes less reliable as the video progresses. It captures the target only in the first frame, while the target may later rotate, deform, change illumination, change scale, or become temporarily occluded. The current search crop can also be unreliable: the object may be blurred, partly hidden, or surrounded by visually similar distractors. In such cases, a tracker needs a mechanism that can dynamically encode the most relevant appearance cues from previous frames instead of relying only on the first template.

Second, tracking is inherently temporal, a simple example can make it clear. Imagine looking at a patch of grass and trying to identify a camouflaged snake, in that one image, the snake may be almost indistinguishable from the background, however, once the video starts playing, coherent movement across consecutive frames can make the same object visible. The object has not necessarily changed its appearance, but the temporal context makes it easier to separate from the background. Additionally, a memory state does not need to store only appearance; it can potentially encode motion cues, learn information about recent target movement: direction, speed, and stable object identity across frames. This can be useful when the current frame is ambiguous, because the previous trajectory and recent observations can provide additional evidence about where the target is likely to be.

A simple way to use temporal information would be to provide the transformer with tokens from previous frames together with the current search crop. However, this quickly leads to a temporal token explosion. For example, if one search crop contains 256 tokens, then 20 previous search crops already contain 5120 tokens before adding template tokens. Even if previous-frame key-value representations are cached, the current frame must still attend to a growing set of stored tokens, increasing both memory usage and attention cost.

Therefore, instead of storing all previous visual tokens, an efficient tracker requires a reliable mechanism for dynamically maintaining a compressed hidden state. Such a state should preserve useful temporal cues from previous frames while keeping computation

bounded as the sequence length increases.

2 OTrack Baseline

2.1 OTrack Tracker

OTrack is a real-time single-object visual tracker introduced by Ye et al. in *Joint Feature Learning and Relation Modeling for Tracking: A One-Stream Framework*, accepted at ECCV 2022 [13]. It is used as the baseline tracker in this thesis because it provides a strong transformer-based architecture while keeping the tracking pipeline relatively simple.

During inference, OTrack processes one search frame at a time. It receives a fixed template crop from the first annotated frame and a search crop from the current frame. The tracker localizes the target inside the search crop, and the predicted bounding box is then used to define the search region for the next frame.

The main architectural idea of OTrack is its one-stream design. Instead of extracting template and search features separately and then applying a separate relation module, OTrack converts both crops into tokens and processes them jointly in a single Vision Transformer (ViT) backbone. As a result, feature extraction and template-search relation modeling are performed inside the same backbone. Self-attention over the combined token sequence allows template and search tokens to interact throughout the network, which helps the model learn target-oriented search features [13].

After the ViT backbone, OTrack uses a lightweight fully convolutional prediction head. The final search tokens are reshaped into a two-dimensional spatial feature map and passed to this head. The head has three fully convolutional output branches, which predict a classification score map, a local offset map, and a normalized bounding-box size map. The position with the highest classification score is selected as the target position. The backbone is initialized from a pretrained ViT model.

OSTrack also supports candidate elimination, which progressively removes less relevant search tokens to reduce computation; this mechanism is described separately in Section 2.3.

2.2 ViT Backbone

OSTrack is built on a pretrained Vision Transformer (ViT) backbone. Most of the modifications in this thesis are focused on the ViT backbone. ViT is a powerful image-processing architecture that has shown strong performance in computer vision and, with large-scale pretraining, can match or outperform state-of-the-art convolutional networks on several image-recognition benchmarks. ViT represents an image as a sequence of patch tokens and processes this sequence using transformer encoder layers [19]. In OSTrack, this structure is adapted to tracking by applying patch embedding to both the template crop and the search crop, and then processing the resulting tokens jointly in the same backbone [13].

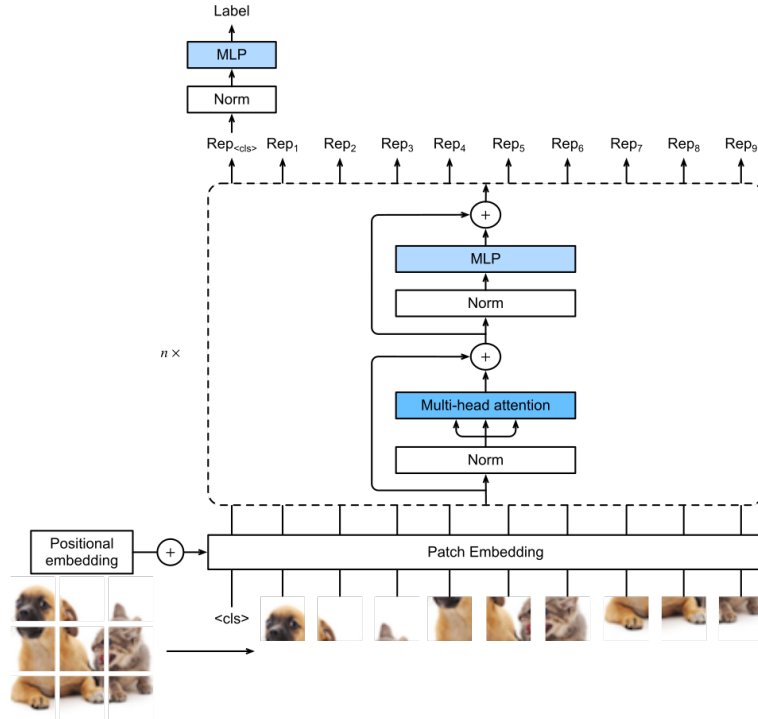


Figure 2.1: Vision Transformer architecture. Source: ‘Vision Transformer.svg’ by Aston Zhang, Zachary C. Lipton, Mu Li, and Alexander J. Smola, Wikimedia Commons, CC BY-SA 4.0 [22].

During the ViT processing of an image crop X , the image is first divided into non-overlapping patches:

$$N = \frac{HW}{P^2}, \quad X_p = [x_1, x_2, \dots, x_N]^\top \in \mathbb{R}^{N \times P^2 C}. \quad (2.1)$$

where:

- $X \in \mathbb{R}^{H \times W \times C}$ is the input image crop;
- H and W are the height and width of the crop;
- C is the number of image channels;
- P is the patch size;
- N is the number of patches;
- $x_i \in \mathbb{R}^{P^2 C}$ is the flattened vector of the i -th patch;
- X_p is the sequence of all flattened patch vectors.

A standard ViT encoder layer applies a pre-normalized multi-head self-attention block followed by a pre-normalized per-token MLP block, with residual connections after both. In the candidate elimination (CE) enabled OTrack backbone, candidate elimination is inserted between the attention block and the MLP block. This gives the following ViTBackbone layer update:

$$U_l = Z_{l-1} + \text{MSA}(\text{LN}(Z_{l-1})), \quad (2.2)$$

$$\tilde{U}_l = \begin{cases} \text{CE}_l(U_l), & \text{if CE pruning is enabled at layer } l, \\ U_l, & \text{otherwise,} \end{cases} \quad (2.3)$$

$$Z_l = \tilde{U}_l + \text{MLP}(\text{LN}(\tilde{U}_l)). \quad (2.4)$$

where:

- l is the transformer layer index;
- Z_{l-1} is the input token sequence to layer l ;
- U_l is the intermediate sequence after the attention block and residual connection;
- $\tilde{U}_l$ is the token sequence after optional CE pruning;
- Z_l is the output token sequence of layer l ;
- $\text{LN}(\cdot)$ denotes Layer Normalization;
- $\text{MSA}(\cdot)$ denotes Multi-Head Self-Attention;
- $\text{CE}_l(\cdot)$ denotes candidate elimination at layer l ;
- $\text{MLP}(\cdot)$ denotes the feed-forward block.

Multi-head self-attention allows each token to exchange information with other to-

kens in the same sequence. Since OTrack processes template and search tokens together, self-attention provides the mechanism through which template-search interaction occurs inside the backbone.

Candidate elimination is specific to OTrack and is not part of the original ViT architecture. It is only present when CE pruning is enabled. The detailed CE mechanism is described separately in Section 2.3.

OTrack applies this ViT backbone to a pair of tracking crops. Let T denote the template crop and S_t denote the search crop at frame t . Their token sequences are denoted by T_0 and $S_{t,0}$, and the tracker processes their concatenation:

$$Z_{t,0}^{\text{track}} = [T_0, S_{t,0}]. \quad (2.5)$$

After L transformer layers, the backbone output is

$$\begin{aligned} Z_{t,L}^{\text{track}} &= \text{ViTBackbone}(Z_{t,0}^{\text{track}}) \\ &= [T_L^{(t)}, S_{t,L}]. \end{aligned} \quad (2.6)$$

where:

- $T_0 \in \mathbb{R}^{N_T \times D}$ is the template-token sequence;
- $S_{t,0} \in \mathbb{R}^{N_S \times D}$ is the search-token sequence for frame t ;
- $[\cdot, \cdot]$ denotes concatenation along the token dimension;
- L is the number of transformer layers;
- $T_L^{(t)}$ is the processed template-token representation;
- $S_{t,L}$ is the processed search-token representation.

The final search-token representation $S_{t,L}$ is reshaped into a spatial feature map and passed to the prediction head.

2.3 Candidate elimination

OTrack introduces candidate elimination (CE) to reduce the number of search tokens processed by the transformer backbone. In a template-search tracker, the search crop contains both the target and many background patches. CE removes search tokens that

are unlikely to correspond to the target, reducing computation while keeping the most relevant candidates [13].

OTrack chooses candidates using attention between the template and search tokens. After the multi-head attention operation in a transformer block, a representative template token is used to score the search tokens. In the standard setting, this representative token is the center template token, because the target is normally centered in the template crop. If this token is denoted by ϕ and the transformer has H attention heads, the score of search token x is computed by averaging attention from ϕ to x over all heads:

$$w_x^\phi = \frac{1}{H} \sum_{h=1}^H w_x^{\phi,h}. \quad (2.7)$$

where:

- w_x^ϕ is the averaged attention score from the representative template token to search token x ;
- ϕ is the representative template token;
- x is a search token;
- H is the number of attention heads;
- h is the attention-head index;
- $w_x^{\phi,h}$ is the attention score from ϕ to x in head h .

Search tokens with the highest scores are kept, while lower-scoring search tokens are removed from the sequence for later transformer layers. If k tokens are kept from n current search tokens, the keep ratio is:

$$k = \text{round}(\rho n). \quad (2.8)$$

where:

- ρ is the keep ratio;
- k is the integer number of search tokens kept after pruning;
- n is the number of current search tokens before pruning.

The kept search-token indices can be written as:

$$I_{\text{keep}} = \text{TopK}(\{w_x^\phi\}_{x \in S}, k). \quad (2.9)$$

where:

- I_{keep} is the set of retained search-token indices;
- $\text{TopK}(\cdot, k)$ selects the k highest-scoring tokens;
- $\{w_x^\phi\}_{x \in S}$ is the set of attention-based scores used for ranking search tokens.

The original order and positions of the remaining search tokens are stored. Before the prediction head, the kept search tokens are restored to their spatial order and the removed positions are zero-padded so that the spatial search feature map can be reconstructed [13].

This matters for MemOSTrack because the proposed model adds memory tokens to the OTrack token sequence. In this implementation, CE is applied only to search tokens. Template tokens are preserved, and memory tokens are also preserved because they are recurrent latent states rather than spatial search candidates. Memory tokens participate in attention, but they are not used to score search candidates and are not removed by CE. Thus, candidate elimination keeps its original role of pruning search-region candidates, while the recurrent memory stream remains available throughout the backbone.

3 Proposed Memory-Augmented OTrack

3.1 Memory-token design motivation

MemOTrack extends OTrack by adding explicit memory tokens to the transformer backbone. These memory tokens are inserted into the same token sequence as the template and search image feature embeddings, so they participate in ViT attention together with the visual tokens. However, unlike template and search tokens, memory tokens are latent hidden-state vectors rather than image patches, and they are carried across frames during tracking.

This design allows the ViT backbone to continuously read from and refine the memory representation through attention. Visual tokens can interact with memory tokens while the current template and search features are processed, and the resulting memory representation can store information that may be useful in later frames.

The architectural changes are mainly applied inside the backbone. The patch embedding and OTrack prediction head are kept close to the baseline implementation, so the model remains an OTrack-style tracker with an added recurrent memory stream.

3.2 Direct transformer-based memory update

The first architectural variant adds memory tokens to the OTrack backbone and lets the ViT layers update them directly. In this version, memory tokens are concatenated with the template and search tokens and are processed by the same transformer layers as the visual tokens. The goal is to allow memory to interact with search and template tokens through the existing attention operations.

The ViT backbone is modified to process transformed template, search, and memory tokens in a single forward pass:

$$[T_{t,l}, S_{t,l}, M_{t,l}] = \text{ViTLayer}_l([T_{t,l-1}, S_{t,l-1}, M_{t,l-1}]). \quad (3.1)$$

Here:

- $T_{t,l}$ denotes the template tokens after layer l while processing frame t ;
- $S_{t,l}$ denotes the search tokens after layer l while processing frame t ;
- $M_{t,l-1}$ is the memory input from the previous transformer layer in the current frame;
- $M_{t,l}$ is the memory output produced directly by the ViT layer;
- $\text{ViTLayer}_l(\cdot)$ denotes transformer layer l .

All token groups have the same embedding dimension D , so they can be concatenated along the token dimension and processed by the same transformer block.

For the first frame of a sequence, no previous memory state exists, so the model uses a learned initial memory state. After each consecutive frame is processed, the backbone returns an updated memory state in the auxiliary output, which is then fed back into the backbone together with the template and the next search crop when processing the next frame.

To distinguish memory tokens from visual tokens, the model adds a learned memory-token type embedding to the memory tokens before they are processed by the backbone. This embedding shifts the memory tokens in feature space and gives the transformer a learned signal that allows the model to distinguish these tokens as memory states rather than image patches.

In the CE-enabled backbone, memory tokens are intentionally excluded from candidate-elimination decisions. First, they are not pruned, because CE removes only search tokens. Second, they are excluded from the template mask used for CE scoring, so they do not directly influence which search tokens are kept or removed. Thus, CE pruning is decided without treating them as either removable candidates or scoring tokens, so memory tokens can influence it only indirectly through their participation in transformer attention.

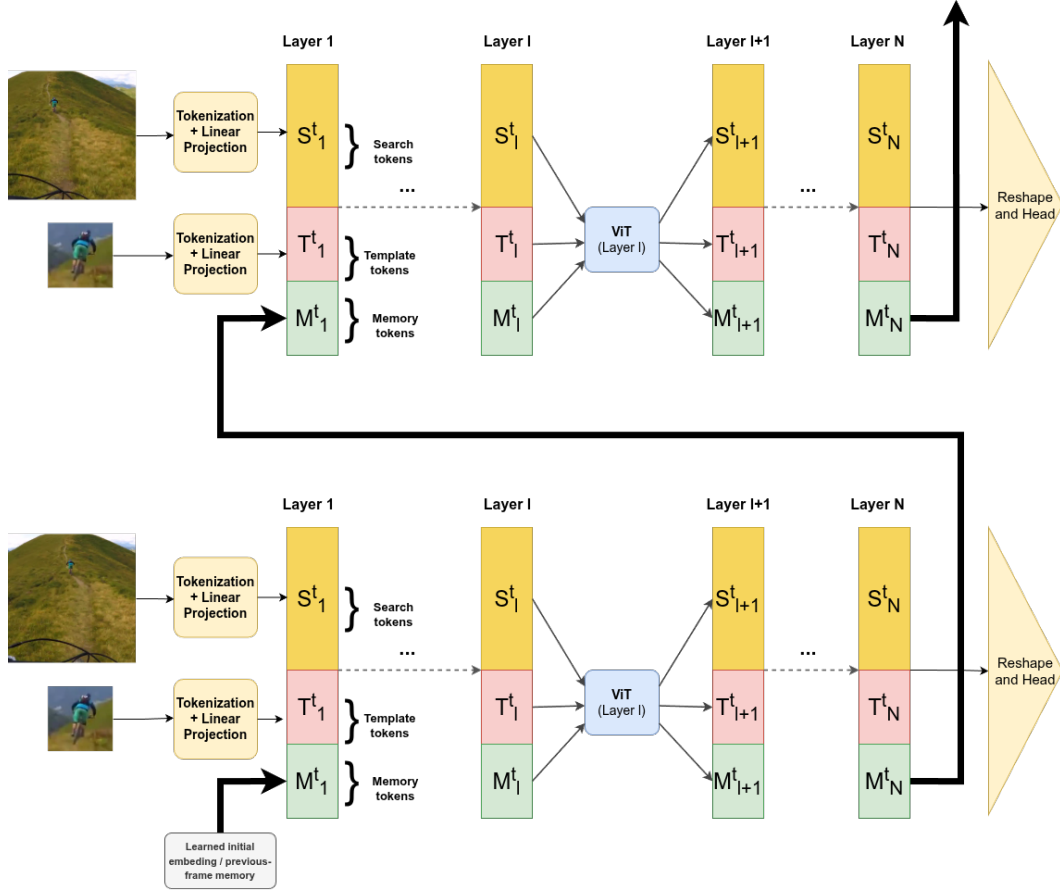


Figure 3.1: Direct transformer-based memory update. The template and search images are first converted into visual tokens by tokenization and linear projection. A learned initial memory state, or the memory state returned from the previous frame, is inserted into the transformer token sequence. Each ViT layer jointly processes template, search, and memory tokens through self-attention. The updated memory state is passed through the backbone and returned for use in the next frame, while the final visual tokens are reshaped and passed to the OSTRack prediction head.

3.3 Limitations of direct memory update

When used for long-term temporal modeling, the direct memory update variant described in Section 3.2 exposes two fundamental limitations. While the standard Vision Transformer architecture is effective for spatial feature extraction and template-search matching, relying on it alone for recurrent memory updates may be insufficient.

The first limitation is the lack of explicit gating for memory preservation. In the direct-update variant, memory tokens are updated directly by the same transformer block. The attention mechanism is effective for deciding which template and search tokens should influence the memory representation, however, it does not provide a dedicated gate that controls how much of the previous memory should be kept and how much new information should be written.

Equation (3.2) provides a residual path that can preserve information from the previous layer:

$$U_l = Z_{l-1} + \text{MSA}(\text{LN}(Z_{l-1})). \quad (3.2)$$

Since Z_{l-1} includes the memory tokens in the direct-update variant, this residual connection passes their previous layer representation forward. While in principle, the model can learn to produce a small attention update, $\text{MSA}(\text{LN}(Z_{l-1}))$, when the memory does not need to change much, this mechanism may be weaker than an explicit update gate. Over long tracking sequences, repeated ungated updates may still gradually alter the memory state and make it less representative of the target.

The second limitation is the susceptibility to gradient instability during Backpropagation Through Time due to an absence of temporal skip connections. In recurrent neural networks, there are often temporal skip connections between the memory representations of the same layer across time. However, in the direct-update design, recurrence is present only in the form of a feedback loop, where the last layer of the current frame is connected to the first layer of the next frame. This means that when unrolling the memory tokens over T frames through an L -layer backbone, the gradient has to travel through all layers of the backbone for each frame.

Consequently, computing how the initial memory state affects the final frame’s loss involves a continuous chain of multiplications of the local gradients across every layer of every frame in the sequence. While this continuous chain isolates the recurrent memory path, the full training gradient also contains paths through search tokens, template tokens, the prediction head, and the backbone parameters.

Suppose we have a 12-layer transformer backbone over a 20-frame sequence; this chain expands to 240 sequential matrix multiplications. This deep computational graph can contribute to error accumulation, causing gradient vanishing or explosion. One way to address this issue is to add explicit temporal skip connections.

To address both memory drift and gradient instability, a more intentional architectural design is required. Specifically, the architecture should decouple the *proposal* of new memory from the actual *update* of the memory state, and it should introduce explicit temporal skip connections to stabilize gradient flow. This motivates the second architectural variant, which replaces the direct update with a two-way gated recurrent mechanism, as detailed in the following section.

3.4 Two-way GRU. GRU-based memory update

To implement this controlled update, the second architectural variant introduces a Gated Recurrent Unit (GRU) mechanism to the memory stream. A GRU is a recurrent module that updates a hidden state using learned gates [20]. These gates control how much of the previous state is preserved and how much new information is accepted. This is highly suitable for memory tokens because the tracker must retain reliable target information while still adapting when the target appearance changes.

In this version, the transformer layer does not directly produce the final memory. Instead, the transformer output is treated as a candidate memory representation:

$$[T_{t,l}, S_{t,l}, M'_{t,l}] = \text{ViTLayer}_l([T_{t,l-1}, S_{t,l-1}, M_{t,l-1}]). \quad (3.3)$$

where:

- $M'_{t,l}$ is the candidate memory representation proposed by the transformer layer;
- $T_{t,l}$ and $S_{t,l}$ are the template and search tokens after layer l at frame t ;
- $M_{t,l-1}$ is the memory from the previous layer in the current frame;
- $\text{ViTLayer}_l(\cdot)$ denotes transformer layer l .

The value $M'_{t,l}$ is the memory suggested by attention at the current frame and layer. It is not accepted directly. Instead, it is passed to a two-stage GRU update. The first GRU combines the candidate memory with the memory from the previous frame at the same layer, creating an explicit temporal skip connection:

$$\tilde{M}_{t,l} = \text{GRU}_1(M'_{t,l}, M_{t-1,l}). \quad (3.4)$$

where:

- $\tilde{M}_{t,l}$ is the intermediate memory after temporal fusion;
- $\text{GRU}_1(\cdot, \cdot)$ is the first GRU update, applied independently to each memory token;
- $M'_{t,l}$ is the current candidate memory and is used as the GRU input;
- $M_{t-1,l}$ is the memory from the previous frame at the same layer and is used as the previous hidden state.

The second GRU combines the intermediate memory with memory from the previous layer in the current frame:

$$M_{t,l} = \text{GRU}_2(\tilde{M}_{t,l}, M_{t,l-1}). \quad (3.5)$$

where:

- $M_{t,l}$ is the final memory output at frame t and layer l ;
- $\text{GRU}_2(\cdot, \cdot)$ is the second GRU update, applied independently to each memory token;
- $\tilde{M}_{t,l}$ is the intermediate memory after temporal fusion and is used as the GRU input;
- $M_{t,l-1}$ is the memory from the previous layer in the current frame and is used as the previous hidden state.

The final output of the layer is:

$$Y_{t,l} = [T_{t,l}, S_{t,l}, M_{t,l}]. \quad (3.6)$$

where:

- $Y_{t,l}$ is the final token sequence output of layer l at frame t ;
- $[\cdot, \cdot, \cdot]$ denotes concatenation along the token dimension;
- $T_{t,l}$, $S_{t,l}$, and $M_{t,l}$ are the template, search, and memory tokens after the layer update.

This recurrent update is used only when previous-frame memory is available. On the first frame, the tracker initializes memory from the learned initial memory state and skips the temporal GRU comparison with a previous frame. The first GRU can be interpreted as temporal fusion. It decides how much memory from frame $t - 1$ should remain when the current candidate memory is introduced. The second GRU can be interpreted as layer-wise fusion. It allows memory from the previous transformer layer to influence the final memory at the current layer, creating a gated depth path for memory tokens.

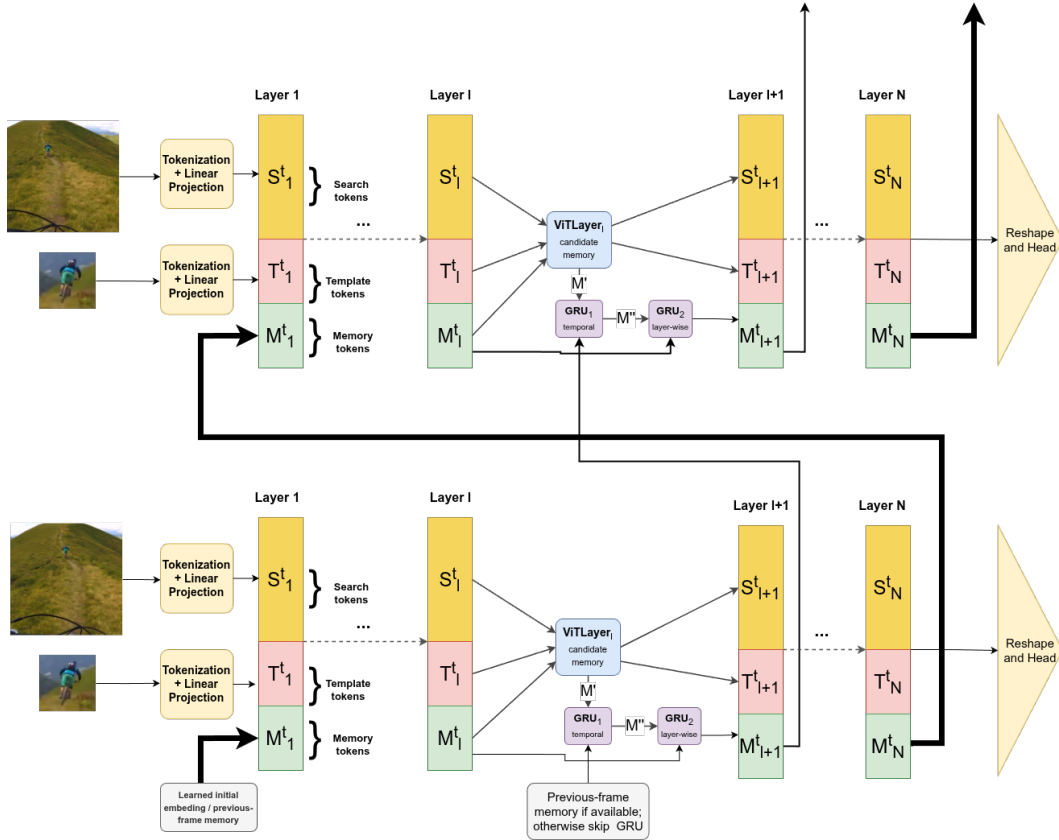


Figure 3.2: GRU-based memory update with temporal and depth connections inside the backbone.

This design was introduced in the later stage of development. The GRU update makes the memory stream explicitly recurrent and gives the model a learned mechanism for balancing preservation and adaptation.

4 Sequence-Based and Inference-Like Training

4.1 Sequence-based training for recurrent memory

Standard OTrack training is based on independent template-search pairs. A template crop is sampled from one frame, a search crop is sampled from another frame in the same video, and the model is trained to predict the target box inside the search crop:

$$\text{sample} = (T, S_t, B_t), \quad \hat{B}_t = f(T, S_t). \quad (4.1)$$

Here:

- sample denotes one pair-based training sample;
- T is the template crop;
- S_t is the search crop at time t ;
- B_t is the ground-truth target box expressed in the coordinate system of the search crop;
- $\hat{B}_t$ is the predicted target box at time t ;
- $f(\cdot)$ is the tracker model.

Because the standard tracker is memory-free, processing each training sample independently is sufficient.

However, for correct MemOTrack training, independent pair-based training is inadequate. A recurrent memory state M_t must evolve over a continuous sequence of frames from the same video. The standard single-step horizon makes it impossible for the network to learn useful long-term temporal dependencies. Conversely, carrying the memory state forward across randomly sampled, unrelated pairs would feed the model incorrect historical context.

Therefore, the sampling pipeline was rewritten to return ordered search-frame se-

quences. In this causal consecutive setting, the sampled frames follow the natural temporal direction of the video. The training sample becomes:

$$\text{sample} = (T, S_1, S_2, \dots, S_N, B_1, B_2, \dots, B_N). \quad (4.2)$$

where:

- sample denotes one sequence-based training sample;
- T is the template crop;
- $S_1, S_2, \dots, S_N$ are ordered search crops from the same video;
- $B_1, B_2, \dots, B_N$ are the corresponding target boxes;
- N is the number of search frames in the sampled sequence.

For each frame in the sequence, the model predicts a bounding box and updates its memory state. The memory output from frame $t - 1$ becomes the input for frame t . Preserving this exact temporal order provides the recurrent memory module with a meaningful path to learn across time.

4.2 Dynamic inference-like search cropping

Sequence-based sampling makes recurrent memory training possible, but it does not by itself remove the mismatch between training and inference. If every search crop is centered using the ground-truth box of the current frame, the tracker receives cleaner inputs than it will receive at test time. The target is usually near the center of the crop, and the model does not experience the consequences of its own previous prediction errors.

In standard ground-truth-centered training, the search crop is generated using the ground-truth box of the current frame:

$$S_t = \text{crop}(I_t, B_t^{\text{gt}}). \quad (4.3)$$

where:

- S_t is the search crop at time t ;
- I_t is the full image at time t ;
- B_t^{gt} is the ground-truth target box at time t .

To simulate imperfect localization, standard training applies artificial random jitter to this ground-truth box. However, even with this jitter, the inputs remain fundamentally

disconnected from the model’s actual performance. Because every crop is anchored to the true target location, the model never experiences the compounding prediction errors that occur during real tracking.

During inference, the tracker does not know the current ground-truth target location. The next search crop must be generated from the previous prediction:

$$S_t = \text{crop}(I_t, \hat{B}_{t-1}). \quad (4.4)$$

where:

- $\hat{B}_{t-1}$ is the predicted bounding box from the previous frame.

This prediction-centered cropping allows localization errors to accumulate. A poor prediction in one frame will shift the target away from the center in the next frame, potentially leading to complete tracking failure. This difference is especially critical for a recurrent tracker, as an inaccurate crop will directly corrupt the memory state passed to subsequent frames.

To close this gap, the final training pipeline introduces inference-like dynamic search cropping. The tracker is initialized using the first-frame ground truth, but all subsequent crops in the sequence are generated dynamically from the model’s predictions. At each step, the model processes the template, the current search crop, and the previous memory state to predict the new target box and update the memory:

$$\begin{aligned} S_t &= \text{crop}(I_t, \hat{B}_{t-1}), \\ \hat{B}_t, M_t &= f(T, S_t, M_{t-1}). \end{aligned} \quad (4.5)$$

where:

- S_t is the search crop generated dynamically for frame t ;
- $\hat{B}_{t-1}$ is the previous predicted box used to center and scale the crop;
- $\hat{B}_t$ is the predicted box at time t ;
- M_t is the updated memory state at time t ;
- T is the template crop;
- M_{t-1} is the memory state from the previous frame;
- $f(\cdot)$ is the MemOSTrack model.

This dynamic loop intentionally makes training more difficult. With ground-truth-

centered crops, the model sees clean examples even if its previous predictions would have been inaccurate. With dynamic cropping, an early localization error can move the next crop away from the target center, which can cause the target to appear partially outside the crop. However, this difficulty is necessary to close the gap between training and inference and provides two major benefits. First, it allows the model to properly learn motion cues, as the displacement of the target within the crop directly reflects its actual movement relative to the previous frame. Second, it gives the model an opportunity to learn how to recover from its own mistakes; by experiencing the consequences of poor predictions during training, the tracker learns to re-center the target and correct tracking drift before it leads to complete failure.

Because dynamic cropping makes optimization more difficult, the final training pipeline used a curriculum training strategy. During the first 40 epochs, search crops were centered using ground-truth target boxes. After this warm-up phase, inference-like dynamic cropping was enabled, so each next crop was generated from the model’s previous prediction. This allowed the model to first learn stable localization before being exposed to prediction drift.

4.3 Truncated Backpropagation Through Time in the training loop

The introduction of memory tokens transforms the tracker into a recurrent model, necessitating careful management of error backpropagation across temporal iterations. Allowing gradients to flow through an entire video sequence is both computationally expensive and highly demanding on GPU memory. Additionally, unrolling the computational graph over long sequences amplifies the risk of gradient instability, such as vanishing or exploding gradients. A standard approach to address this is truncating backpropagation through time. In the provided implementation, the model periodically detaches the memory state from the computation graph after a fixed number of frames (specifically, 8 frames during training). This truncation limits how far the gradients can propagate backward in time, balancing the need to learn temporal dependencies with computational feasibility and training stability.

5 Auxiliary Memory Supervision

5.1 Auxiliary Loss Concept

Adding memory tokens to a strong model does not guarantee that the model will actually use them. OSTRack already possesses a powerful template-search matching pathway. Because the tracker can often solve the training task using template and search tokens alone, the newly introduced memory tokens may receive weak gradients. This creates the risk that the memory tokens can collapse to an uninformative representation or are entirely ignored by the prediction head.

This is a common issue when adding auxiliary modules to strong neural networks: the optimization process often follows the dominant low-loss pathway. If the template is clean and the search crop is well-centered, that pathway may rely entirely on the original OSTRack features, leaving the memory branch with little to no effect on the final prediction.

To address this, this thesis explored auxiliary memory supervision. The goal was to force the memory tokens to encode representations that are directly useful for prediction, rather than remaining passive elements within the token sequence.

5.2 Sampled Discriminative Filter Loss

The first auxiliary loss was designed to explicitly force the memory tokens to encode spatial representations capable of localizing the target. To achieve this, an auxiliary memory head was introduced to the architecture. This lightweight multi-layer perceptron takes only the recurrent memory tokens as input and projects them into a spatial convolutional filter. Rather than allowing the memory tokens to remain abstract, this approach evalu-

ates their discriminative power by testing how well this predicted filter can localize the target across the video sequence.

Let f_t denote the memory filter predicted by the memory head at frame t . To evaluate this filter, a random subset of frames, denoted by the index set S_t , is sampled from the current training sequence. For each sampled frame $j \in S_t$, the filter f_t is applied to the corresponding search region feature map x_j via cross-correlation to produce a spatial response map:

$$s_{t,j} = x_j * f_t. \quad (5.1)$$

where:

- $s_{t,j}$ is the spatial response map produced by filter f_t on frame j ;
- x_j is the search-region feature map for sampled frame j ;
- f_t is the memory filter predicted at frame t ;
- $*$ denotes cross-correlation.

The objective is to minimize the discrepancy between this predicted response map and the ground-truth Gaussian target heatmap y_j . The loss for the filter at frame t is formulated as the mean squared error over the sampled subset, combined with an L_2 regularization penalty on the filter weights:

$$L_t = \frac{1}{|S_t|} \sum_{j \in S_t} \|x_j * f_t - y_j\|_2^2 + \lambda_{\text{reg}} \|f_t\|_2^2. \quad (5.2)$$

Here:

- L_t is the auxiliary loss for the filter predicted at frame t ;
- S_t is the sampled subset of frames used to evaluate filter f_t ;
- $|S_t|$ is the number of sampled frames in that subset;
- j is the sampled-frame index;
- x_j is the search-region feature map for frame j ;
- f_t is the memory filter predicted at frame t ;
- y_j is the ground-truth Gaussian target heatmap for frame j ;
- λ_{reg} is a hyperparameter controlling the strength of the weight regularization.

The total auxiliary loss is the average of L_t across all frames in the sequence that produce a memory filter.

A critical design choice in this algorithm is the treatment of the search features x_j . During the loss computation, these features are explicitly detached from the computational graph. If the search features were not detached, gradients from this auxiliary loss would flow backward through them and into the ViT backbone for the sampled frames. That would force the backbone to modify its general visual representations to accommodate the memory loss. By detaching the features, the optimization gradients are forced to propagate exclusively backward through the predicted filter f_t , the auxiliary memory head, and into the recurrent memory tokens. This ensures that the loss strictly supervises the memory pathway without inadvertently altering the primary feature extractor.

The primary strength of this loss is that it evaluates the memory representation in a direct, tracking-oriented manner: the memory state must generate a filter that yields a high response at the target location and a low response on the background, even when applied to frames from different temporal steps.

However, this approach presents two significant limitations. First, because the supervision relies on a randomly sampled subset of frames (S_t), the training signal can exhibit high variance and noise. Second, the computational complexity scales poorly. Evaluating every predicted filter against every other frame in a sequence of length K requires $O(K^2)$ cross-correlation operations. While random sampling reduces this to $O(K \cdot |S_t|)$, it remains computationally heavy. These limitations motivated the development of a more efficient, sequence-level approach, detailed in the following section.

5.3 DiMP Target-Match Loss

The second auxiliary loss is the DiMP-style target-match loss. Instead of directly testing every predicted memory filter on sampled features from previous frames, this loss first constructs a stronger sequence-level target filter and then trains each predicted memory filter to match it.

This idea is inspired by DiMP [15], where tracking is formulated as learning a target-specific discriminative model from training samples. DiMP showed that online target model prediction can use both target and background information, instead of only matching a target template [15].

The proposed DiMP-style target-match loss is not a reimplementation of the full DiMP tracker. Instead, it borrows the idea of fitting a discriminative target filter from a sequence of search features. The fitted filter is then used as a detached reference filter for the memory branch. This differs from original DiMP, where the optimized filter is the actual target classifier used for tracking. The simplification is intentional: the goal is not to replace OTrack’s prediction head, but to provide direct supervision that encourages the added memory tokens to encode information useful for target discrimination.

To construct the reference filter, the method first collects all available detached search features, $x_1, \dots, x_K$, and their corresponding ground-truth heatmaps, $y_1, \dots, y_K$. An initial filter is formed by taking the mean of the detached memory filters predicted by the auxiliary memory head across the sequence. A sequence-level target filter, f^* , is then fitted by minimizing the following least-squares objective:

$$L(f) = \text{mean}_j (\|x_j * f - y_j\|_2^2) + \lambda_{\text{reg}} \text{mean} (\|f\|_2^2). \quad (5.3)$$

Here:

- $L(f)$ is the least-squares objective for fitting a target filter;
- f is the candidate discriminative filter being optimized;
- x_j is the detached search feature map for frame j ;
- y_j is the corresponding ground-truth heatmap;
- $*$ denotes cross-correlation;
- λ_{reg} is the regularization strength;
- $\text{mean}_j(\cdot)$ denotes averaging over the evaluated frames;
- $\text{mean}(\cdot)$ denotes averaging over the remaining batch, spatial, or filter dimensions;
- f^* is the filter that best matches the whole training sequence under this least-squares objective.

The implementation solves this using a steepest-descent solver with an analytic step length.

After computing the target filter, the memory filters predicted by the memory head are trained to match it. The final auxiliary loss is the mean squared difference between each predicted filter and the optimized target filter:

$$L_{\text{match}} = w_{\text{match}} \frac{1}{K} \sum_{i=1}^K \text{mean} (\|f_i - f^*\|_2^2). \quad (5.4)$$

where:

- L_{match} is the target-match auxiliary loss;
- w_{match} is FILTER_MATCH_WEIGHT;
- K is the number of predicted memory filters;
- i indexes the predicted memory filters;
- f_i is the predicted memory filter from frame i ;
- f^* is the optimized sequence-level target filter.

This loss has several advantages over the sampled discriminative filter loss described in Section 5.2. It provides the memory branch with a more stable target, because f^* is estimated from the full sequence rather than a small random sample. It trades the computation of cross-correlation on every frame during loss evaluation for the upfront optimization of a single sequence-level correlation filter. While running the inner optimization solver is a heavier operation, it ultimately becomes faster than the quadratic complexity emerging from recomputing cross-correlations between every predicted filter and all previous frames. Furthermore, the loss encourages temporal consistency, as all memory filters are pulled toward the same sequence-level target model.

Despite its computational advantages, the target-match loss introduces notable trade-offs. Primarily, it relies heavily on the quality of the reference filter. If the inner optimization yields an inaccurate filter f^* , such as during heavy occlusion, the memory tokens are penalized for imitating a flawed model. Furthermore, this supervision is highly prescriptive. By forcing memory tokens to project into a spatial correlation filter, it restricts the transformer from freely encoding other potentially useful temporal cues, such as motion or abstract relations.

Because of the architectural complexity and constrained nature of explicit auxiliary losses, this thesis also explored a complementary, implicit regularization strategy: Template Blurring, detailed in the following section.

5.4 Template Blurring

To complement the explicit auxiliary losses, this thesis also explores template blurring as an implicit memory-supervision strategy. Because the baseline OTrack architecture already possesses a highly effective template-search matching pathway, the model can often solve the training task without utilizing the newly introduced memory tokens. Template blurring artificially degrades the reliability of the initial template and may encourage the tracker to use the recurrent memory tokens as an alternative source of target appearance information.

Let z denote the original template and $\tilde{z}$ the corrupted template. Two corruption modes were considered: a blur mode ($\tilde{z} = \text{Blur}(z)$) using repeated average pooling, and a zero-masking mode ($\tilde{z} = 0$). The blur mode is less aggressive, removing high-frequency appearance details while preserving coarse structural cues. However, because the model might still recover useful information from a blurred template, zero-masking was also introduced. By completely removing the template’s visual features, zero-masking creates maximum pressure on the network to extract and utilize the historical target information stored in the memory tokens.

During training, template corruption is applied randomly to a subset of frames within the sampled sequence. A critical design choice is that the first few search frames are strictly excluded from corruption. This allows the tracker to process clean frames initially and build a reliable memory state before template corruption is introduced. Consequently, this mechanism is only applied to sufficiently long training sequences to ensure a proper balance between clean and corrupted inputs.

Template corruption is strictly a training-time regularization technique; during inference, the tracker always receives the clean initial template.

The primary advantage of template blurring is its architectural simplicity. It requires no additional loss heads, inner optimization solvers, or reference filters. The tracker is optimized entirely through the standard localization losses, while the input corruption naturally shifts the network’s internal reliance toward the memory stream. Furthermore, because it alters the actual tracking pathway rather than just an intermediate representation, improved performance on corrupted frames suggests that the memory tokens may

provide target appearance cues.

As a result, template blurring serves as a controlled training condition that weakens the dominant template matching, offering a simple and implicit alternative to explicit auxiliary losses.

6 Experimental Setup

6.1 Dataset and protocol

The experiments are performed on the GOT-10k dataset [17], a generic object tracking benchmark comprising over 10,000 video segments and 1.5 million labeled bounding boxes. Because the test set annotations are withheld, final evaluation is performed by submitting predictions to the official benchmark server. To ensure a fair comparison, training is conducted exclusively on the GOT-10k training split, without incorporating any additional tracking datasets.

6.2 Model configuration

As a baseline, the OTrack model with a 256-pixel search crop and Candidate Elimination (CE) pruning was chosen. Following the original OTrack configuration, the ViT backbone is initialized from an MAE-pretrained checkpoint [21].

The proposed MemOTrack configuration builds upon this baseline with the following specifications. The model processes a 128×128 pixel template crop (64 template tokens) and a 256×256 pixel search crop (256 search tokens). To this visual sequence, 64 memory tokens are added, bringing the initial sequence length to 384 tokens. The memory tokens are updated inside the transformer backbone via the proposed two-stage GRU recurrent mechanism.

The training pipeline employs a causal consecutive sequence sampler under a two-stage curriculum. To stabilize early optimization, the first 40 epochs utilize ground-truth-centered crops with random jitter over 20-frame rollouts. After this warm-up stage, inference-like dynamic cropping is fully enabled, and the rollout length is reduced to 15

frames.

6.3 Evaluation protocol

The tracker is evaluated on the GOT-10k test set using Average Overlap (AO), Success Rate at 0.50 (SR0.50), and Success Rate at 0.75 (SR0.75), as defined in Section 1.4.

The comparison is made against reported OTrack-256 + CE and OTrack-384 + CE baselines. The 256-pixel baseline is the closest architectural comparison because it uses the identical template and search crop resolution, while the 384-pixel baseline is included as a stronger, high-resolution reference.

Because the proposed MemOTrack model introduces several changes simultaneously, the final result should not be interpreted as identifying a single cause. A comprehensive ablation study separating the individual effects of memory tokens, dynamic cropping, GRU configurations, and auxiliary loss weights is left for future work.

7 Results and Discussion

7.1 Quantitative Results

The final measured result for MemOTrack-256 + CE is shown in Table 7.1. The model achieved AO 0.729, SR0.50 0.823, and SR0.75 0.693. The reported OTrack-256 + CE baseline achieves AO 0.710, SR0.50 0.804, and SR0.75 0.682. The stronger high-resolution OTrack-384 + CE baseline achieves AO 0.737, SR0.50 0.832, and SR0.75 0.708 [13].

Table 7.1: Final quantitative comparison on GOT-10k.

Method	AO	SR0.50	SR0.75
OTrack-256 + CE	0.710	0.804	0.682
MemOTrack-256 + CE	0.729	0.823	0.693
OTrack-384 + CE	0.737	0.832	0.708

The final MemOTrack configuration improves upon the reported same-resolution OTrack-256 + CE baseline across all three metrics, achieving gains of +0.019 in AO, +0.019 in SR0.50, and +0.011 in SR0.75. This shows that the complete MemOTrack training pipeline can improve tracking accuracy at the same search resolution.

The result remains slightly below the reported OTrack-384 + CE baseline. This is expected, because the 384-resolution baseline processes a larger template and search region, giving it access to finer spatial detail. MemOTrack-256 closes part of the gap to this stronger high-resolution model, but it does not surpass it.

To contextualize the accuracy comparison, Table 7.2 reports the measured inference speed on an NVIDIA RTX A6000 GPU. The table also includes the initial number of input

tokens processed by the backbone. With a patch size of 16×16 , OTrack-256 uses 64 template tokens and 256 search tokens, resulting in 320 initial tokens. MemOTrack-256 adds 64 recurrent memory tokens, increasing the initial sequence length to 384 tokens. In contrast, OTrack-384 uses a larger template and search region, resulting in 720 initial tokens.

Table 7.2: Input-token budget and inference speed on NVIDIA RTX A6000.

Method	Initial tokens	FPS
OTrack-256 + CE	320	131.46
MemOTrack-256 + CE	384	97.73
OTrack-384 + CE	720	72.85

The measured inference speeds confirm that incorporating recurrent memory introduces a measurable computational overhead. MemOTrack-256 + CE runs at 97.73 FPS, compared with 131.46 FPS for the direct OTrack-256 + CE baseline. This slowdown is expected because the memory tokens increase the sequence length processed by the transformer backbone.

At the same time, MemOTrack-256 + CE remains substantially faster than the high-resolution OTrack-384 + CE baseline, which runs at 72.85 FPS. This is important because ViT-based trackers are sensitive to sequence length, especially in the self-attention layers where the cost grows quadratically with the number of tokens. Although runtime is not determined by token count alone, the measured FPS supports the intended trade-off: MemOTrack-256 improves over the same-resolution OTrack-256 baseline while remaining more efficient than simply increasing the tracking resolution to 384. The remaining accuracy gap to OTrack-384 + CE is therefore consistent with the larger spatial token budget and higher computational cost of the high-resolution model.

7.2 Training Dynamics

To understand how the model adapts during training, the learning dynamics were analyzed throughout the training process. Figure 7.1 and Figure 7.2 illustrate the training

and validation metrics, respectively, which reflect the distinct phases of the training strategy.

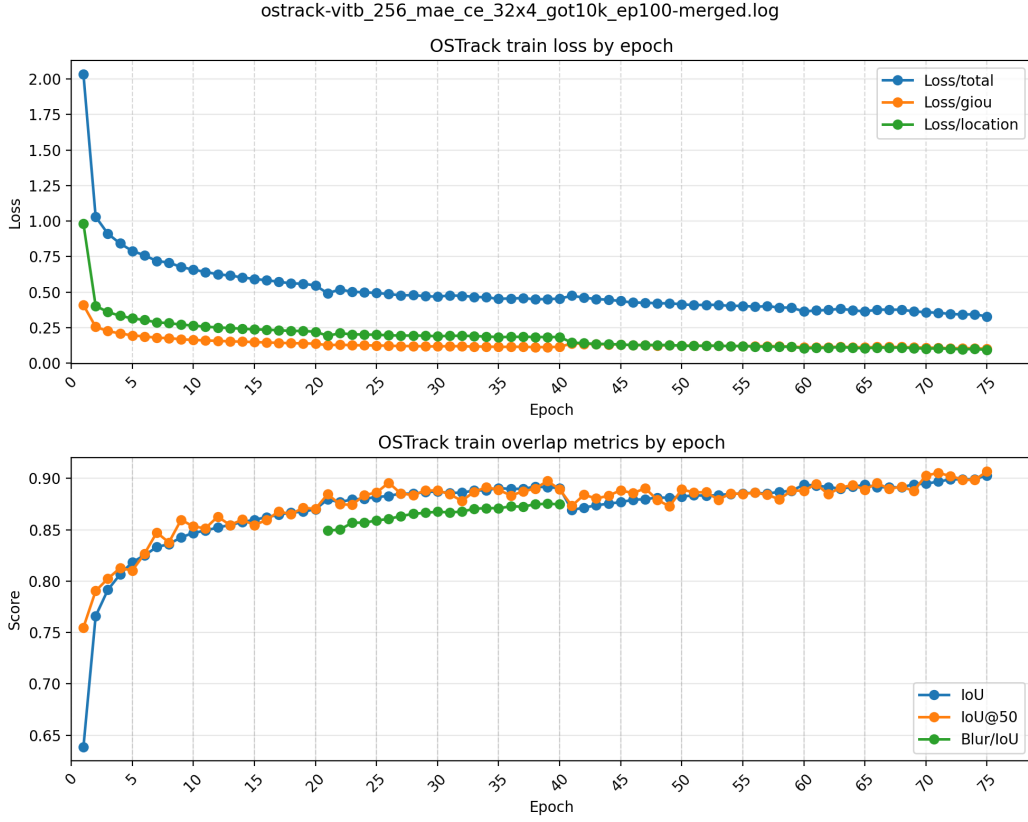


Figure 7.1: Training loss and overlap metrics over the 75 recorded epochs.

The training curve shows rapid early convergence during the initial warm-up phase, in which the model is trained with standard ground-truth-centered crops. In this period, the total training loss decreases sharply from 2.033 to 0.547, while the training IoU climbs to approximately 0.870.

Between epochs 21 and 40, template blurring was introduced to the training pipeline. During this phase, the standard training IoU remained stable and high. More importantly, the model’s performance on specifically corrupted templates, tracked via the separate Blur/IoU metric, steadily improved from 0.849 at epoch 21 to a peak of 0.875 at epoch 39. This upward trend indicates that the network successfully adapted to replace the degraded template with memory information.

At epoch 41, template blurring was disabled, and the training pipeline transitioned to the inference-like dynamic rollout setting. This shift intentionally introduced a significantly harder optimization objective, as the model was now forced to handle its own pre-

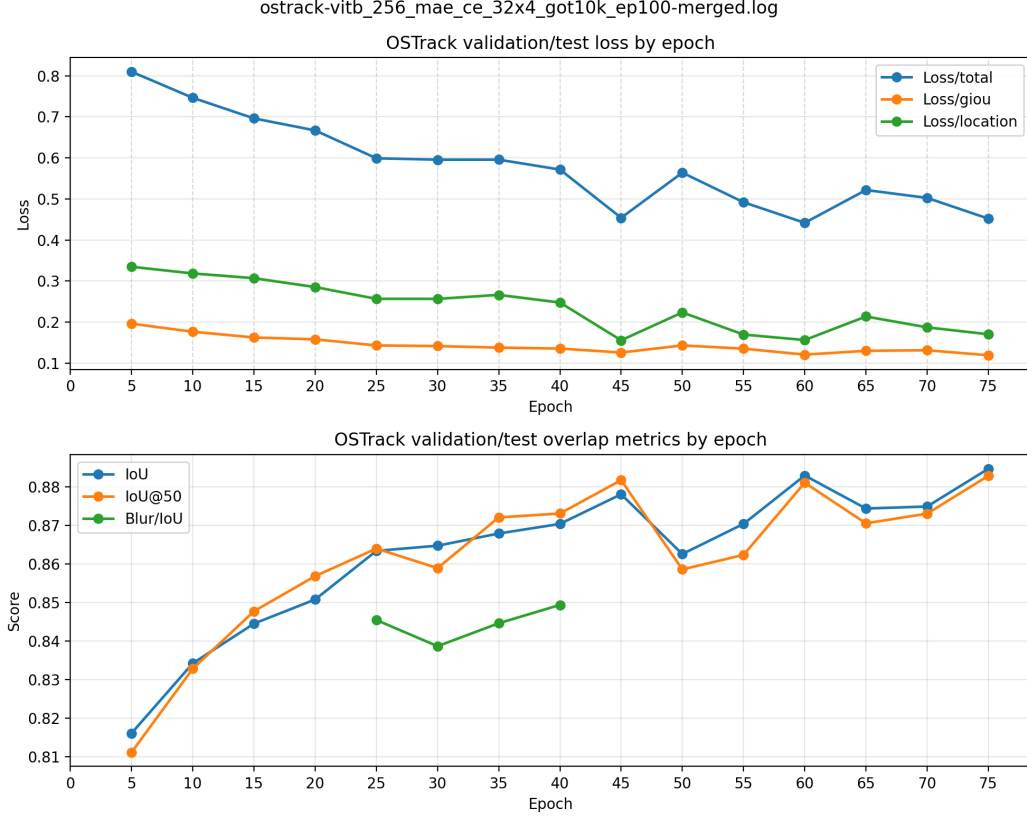


Figure 7.2: Validation loss and overlap metrics recorded every five epochs.

diction drift across frames. This increased difficulty is immediately visible in the graphs: the GIoU loss spikes from 0.114 to 0.139, and the training IoU drops from 0.891 to 0.869. However, the training metrics recovered in subsequent epochs. A scheduled learning rate decay at epoch 60 resulted in a further, immediate reduction in training loss and a corresponding jump in training accuracy. The validation metrics (Figure 7.2) support the same interpretation. The validation loss reached its lowest point of 0.442 at epoch 60. Although the internal validation IoU continued to rise slightly in later epochs (reaching 0.885 at epoch 75). the epoch-60 checkpoint exhibited the best overall validation loss profile. For this reason, the epoch-60 checkpoint was selected for the official GOT-10k test evaluation reported in Table 7.1.

Overall, the training dynamics show that the model first learns stable localization under clean conditions, then adapts to corrupted-template training, and finally becomes robust to inference-like dynamic cropping.

7.3 Limitations and Future Work

Because the final model combines several changes: memory tokens, GRU-based updates, sequence-based training, dynamic cropping, and auxiliary supervision strategies, a full ablation study would be required to separate the contribution of each component. Since these changes were not tested separately, the final result cannot show how much each component contributed to the improvement over the OTrack-256 + CE baseline.

Future work should therefore evaluate these components in separate test runs. In particular, the effects of memory tokens, the GRU update, the recurrent training loop, dynamic cropping, auxiliary memory losses, and template blurring should be studied independently. This would make it possible to understand which parts of the pipeline are most important and which parts mainly add complexity.

Additionally, future work should also analyze the computational trade-off introduced by the memory mechanism. Since memory tokens and GRU updates increase the amount of computation compared with the original OTrack-256 + CE baseline. Testing different numbers of memory tokens, shorter and longer sequence rollouts, and simpler memory-update variants would help determine whether similar accuracy can be achieved with lower computational cost.

Conclusion

This thesis investigated the integration of recurrent memory tokens into OSTRack, a one-stream transformer tracker for single-object tracking. The proposed MemOSTrack model extends the original template-search token sequence with memory tokens that are updated through the ViT backbone and a two-stage GRU mechanism. The goal of this design is to provide the tracker with an explicit recurrent state while preserving the main structure of the OSTRack baseline.

Since recurrent memory cannot be trained effectively with independent frame pairs, the training pipeline was also modified. The original pair-based sampling was replaced with ordered sequence-based training, and inference-like dynamic search cropping was introduced so that later search crops depend on the model’s own previous predictions. Auxiliary memory supervision and template blurring were also explored as ways to encourage the model to use the newly introduced memory stream.

On the GOT-10k benchmark, the final MemOSTrack-256 + CE model achieved an AO of 0.729, an SR0.50 of 0.823, and an SR0.75 of 0.693. This improves over the reported same-resolution OSTRack-256 + CE baseline, while remaining below more expensive high-resolution OSTRack-384 + CE baseline.

The main direction for future work is therefore a systematic ablation study. The effects of memory tokens, the GRU update, the sequence-based training loop, dynamic cropping, auxiliary memory losses, and template blurring should be evaluated in separate test runs. Such experiments would make it possible to determine which parts of the proposed pipeline are most important and how the method can be simplified or improved further.

Although this thesis evaluated the method in the context of Single-Object Tracking,

the proposed combination of recurrent memory tokens, layer-wise GRU updates, and dynamic sequence training may also be relevant to broader video-processing tasks that require compact long-term temporal modeling in Vision Transformers.

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Summary

Extending OTrack with GRU-Based Memory Tokens for Visual Object Tracking

This thesis studies whether a transformer-based visual tracker can benefit from an explicit recurrent memory mechanism. OTrack is selected as the baseline because it is a strong one-stream tracker that jointly processes template and search tokens inside a single Vision Transformer backbone. The proposed MemOTrack model extends this token sequence with recurrent memory tokens.

Two memory-update designs are discussed. The first design lets the transformer update the memory tokens directly. The final design instead treats the transformer output as a candidate memory representation and updates it with a two-stage GRU mechanism. This update combines information from the previous frame and the previous transformer layer, giving the memory stream both temporal and layer-wise connections.

A major part of the work is the training pipeline. Standard OTrack training uses independent template-search pairs, which is not sufficient for learning recurrent memory. Therefore, the sampler was rewritten to return ordered video sequences. The final pipeline also uses dynamic search cropping, where each next search crop is generated from the previous model prediction rather than from the current ground-truth box. This makes training closer to real tracking inference.

The thesis also discusses memory loss heads and template blurring. Memory loss heads were introduced to prevent the model from ignoring memory tokens. Template blurring was explored to reduce over-reliance on the clean initial template and encourage the use of temporal memory.

The final evaluation shows that MemOTrack-256 + CE achieved AO 0.729, SR0.50 0.823, and SR0.75 0.693. These results are higher than the reported OTrack-256 + CE baseline, but lower than the reported OTrack-384 + CE baseline. Further ablation studies are required to separate the effects of the individual components in the final pipeline.

Abbreviations

Table 7.3: Abbreviations.

Abbreviation English term		Explanation
AO	Average Overlap	Average IoU over evaluated frames
BPTT	Backpropagation	Training procedure for recurrent models
	Through Time	unrolled over sequences
CE	Candidate Elimination	OSTrack efficiency module that removes unlikely tokens
GOT-10k	Generic Object Tracking benchmark	Large-scale tracking benchmark
GRU	Gated Recurrent Unit	Recurrent neural network unit with gates
IoU	Intersection over Union	Overlap between predicted and ground-truth boxes
MAE	Masked Autoencoder	Self-supervised vision pretraining method
SOT	Single-Object Tracking	Tracking one specified object through a video
SR	Success Rate	Fraction of frames above an IoU threshold
ViT	Vision Transformer	Transformer model applied to image patches